Car Performance Analysis

Presented By:

Jitu Prajapati

Context : CAR PERFOMANCE OVERVIEW

Problem statement

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- Increasing car data makes it difficult to analyze performance, fuel efficiency, and transmission manually
- Different car brands have varying features, making comparison challenging
- This project uses data visualization to simplify data and provide clear insights

Overview

- This project analyzes car data using data visualization techniques
- The dataset includes features like horsepower, MPG, engine size, fuel type, and transmission
- Charts are used to compare brands and analyze performance
- The goal is to simplify data and provide clear insights

Who will use

Who Will Use

- Car manufacturers to analyze and compare performance

- Buyers to choose cars based on features and efficiency

- Analysts to study trends and insights

DATA OVERVIEW

Data Source:
Kaggle

Dataset Size:
Total Records: 300 rows and 10 columns

Columns Included:
Car Model – Name of the car
Brand – Manufacturer of the car
Engine Size – Size of engine (in liters)
Horsepower – Power output of the car
MPG – Fuel efficiency (miles per gallon)
Weight – Weight of the car
Acceleration – Speed of 0–100 km/h
Cylinders – Number of engine cylinders
Fuel Type – Petrol, Diesel, Electric
Transmission – Manual, Automatic, CVT

DASHBOARD DESIGN

- **Bar Chart:** Comparison of Average Horsepower across different car brands

- **Clustered Column Chart:** Comparison of fuel efficiency (MPG) by car brand

- **Pie Chart:** Distribution of cars based on fuel type

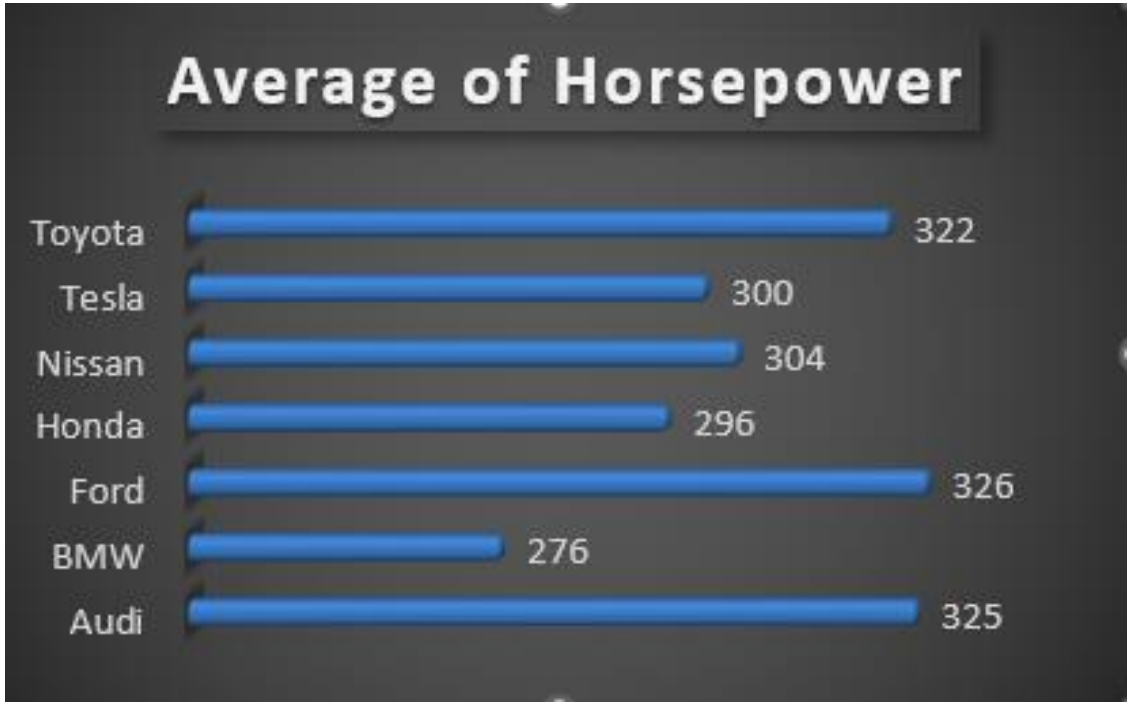
- **Doughnut Chart:** Distribution of cars based on transmission type

- **Bar Chart:** Comparison of average engine size across brands

- **Line with Markers Chart:** Trend of avrage acceleration across car brands

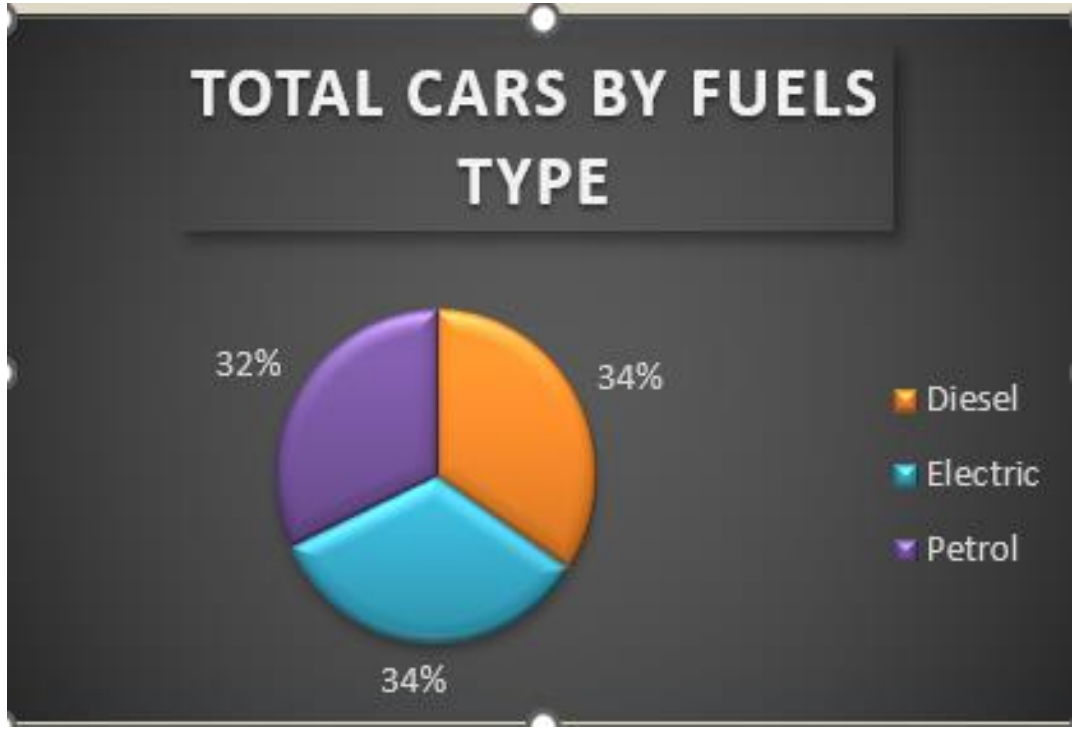
KEY INIGHTS

The bar chart shows that Ford has the highest average horsepower at **326 HP**, followed by Audi (**325 HP**) and Toyota (**322 HP**). Nissan and Honda show moderate values, while BMW has the lowest at **276 HP**, indicating comparatively lower performance.



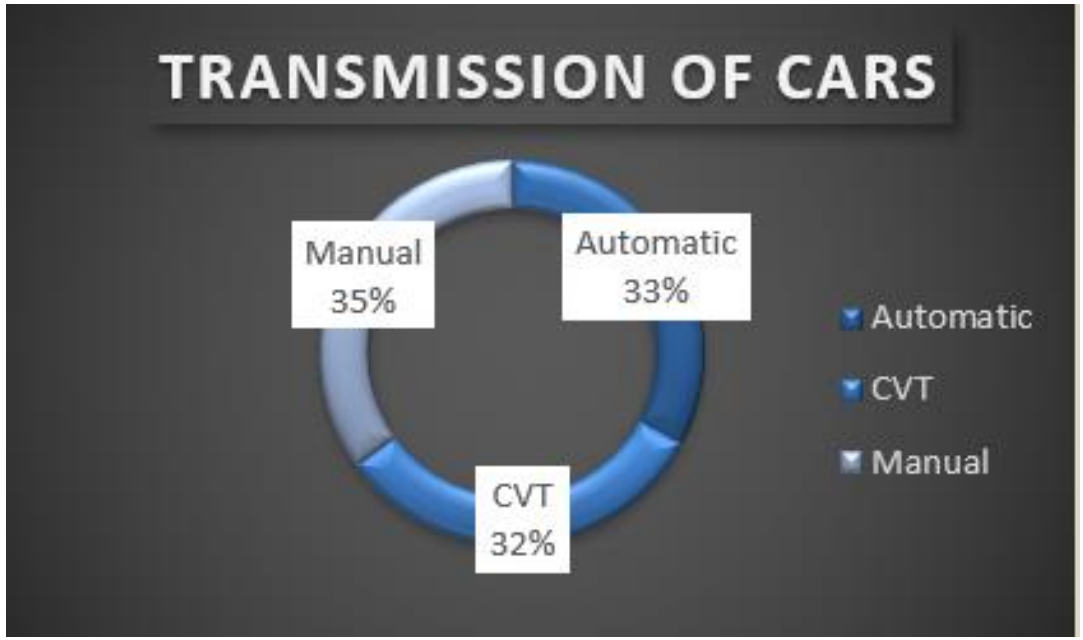
KEY INSIGHTS

The pie chart shows that petrol cars have the highest share at around **34%**, followed by electric cars at approximately **33%**, while diesel cars account for about **32%**. This indicates that petrol cars dominate the dataset, with electric vehicles also having a strong presence..



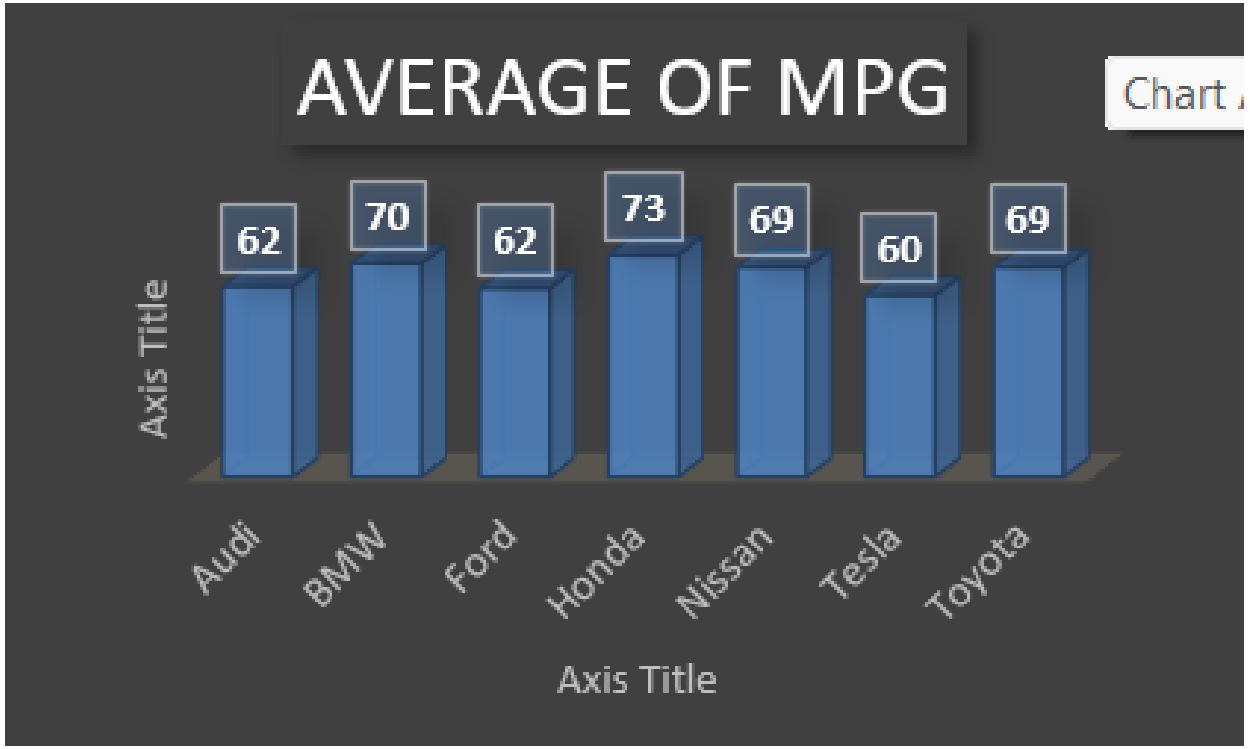
KEY INSIGHTS

The doughnut chart shows manual transmission at **35%**, automatic at **33%**, and CVT at **32%**. The distribution is almost equal, showing balanced usage.



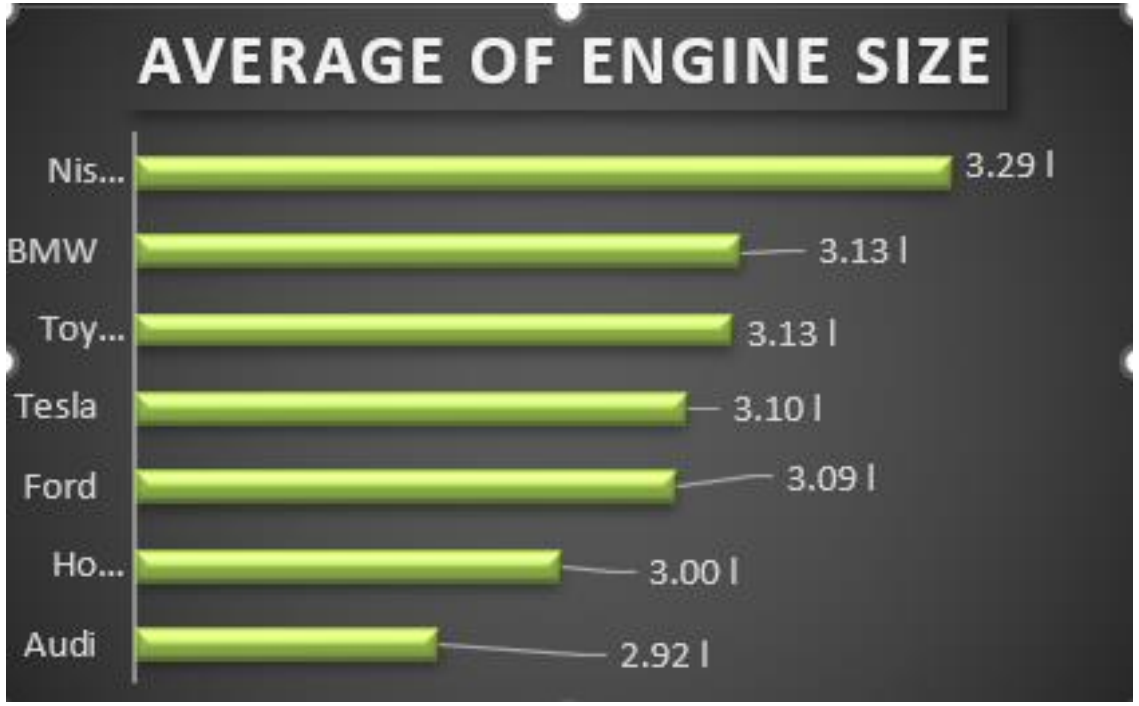
KEY INSIGHTS

The column chart shows that Honda has the highest fuel efficiency at 73 MPG, followed by BMW (70 MPG). Nissan and Toyota have moderate values around 69 MPG, while Ford and Audi are at 62 MPG, and Tesla is lowest at 60 MPG.



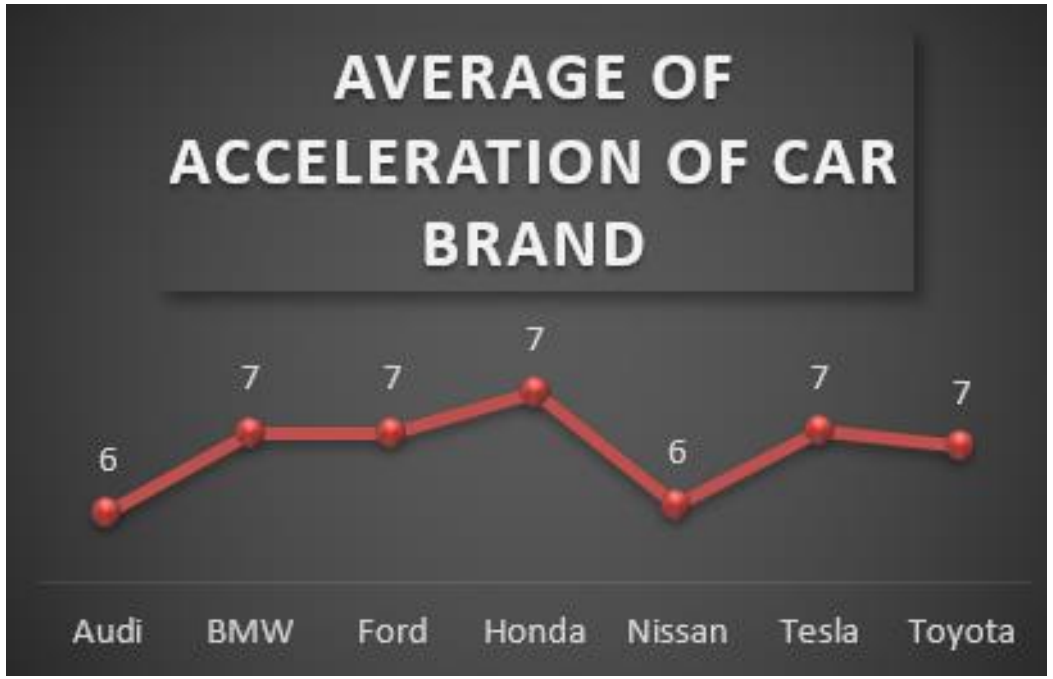
KEY INSIGHTS

The bar chart shows most brands have engine size around **3.0 L**, with Nissan slightly higher (~**3.2 L**) and Audi slightly lower (~**2.8 L**), indicating small variation.



KEY INSIGHTS

The line chart shows acceleration is consistent across brands, ranging between **6 to 7 seconds**, with only minor difference



Future Enhancements

- **Future Enhancements**
- • Add real-world data for better accuracy
- • Include more features like price and top speed
- • Improve charts with better interactivity
- • Use real-time data for updated insights
- • Add prediction for future performance

Conclusion

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The dashboard analyzes car performance based on parameters like horsepower, MPG, engine size, fuel type, and transmission. It helps compare different brands and identify trends. Using charts and pivot tables makes the data easy to understand. Overall, it supports better analysis and decision-making.